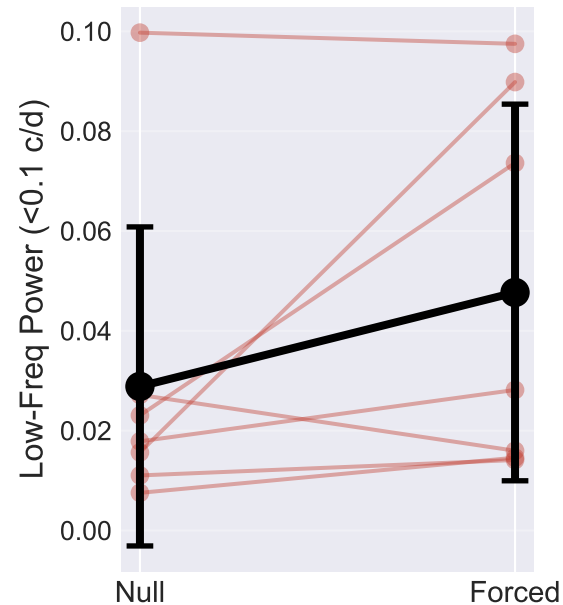


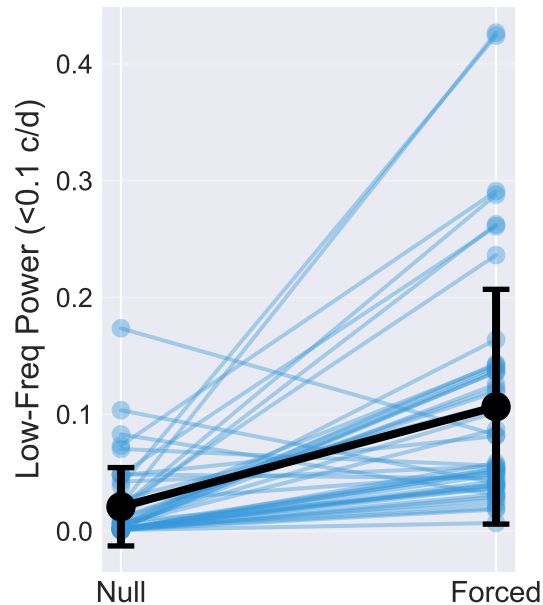
Low freq shift: +0.019(+65.2%)
% outbreaks slowing: 28.6%

COVID (N=7)



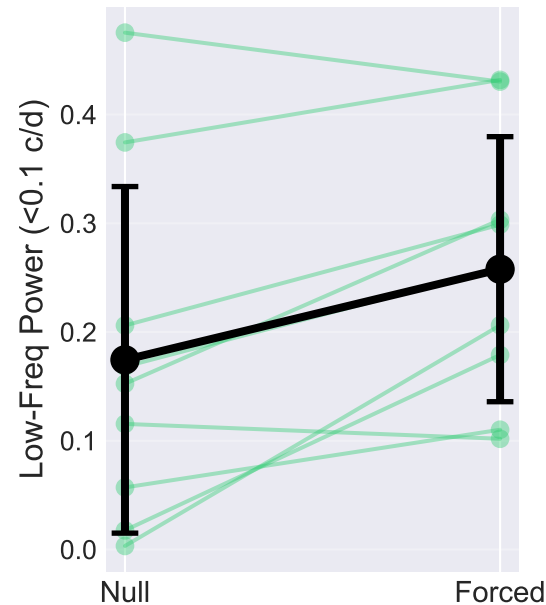
Low freq shift: +0.086(+410.6%)
% outbreaks slowing: 68.1%

Mpox (N=47)



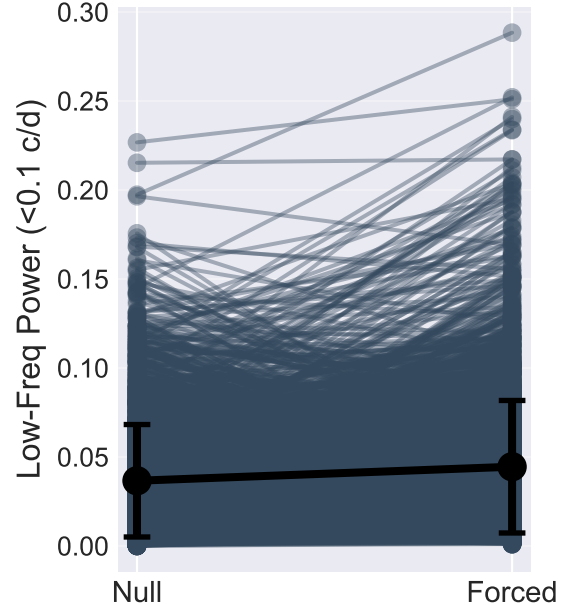
Low freq shift: +0.083(+47.8%)
% outbreaks slowing: 22.2%

Influenza (N=9)

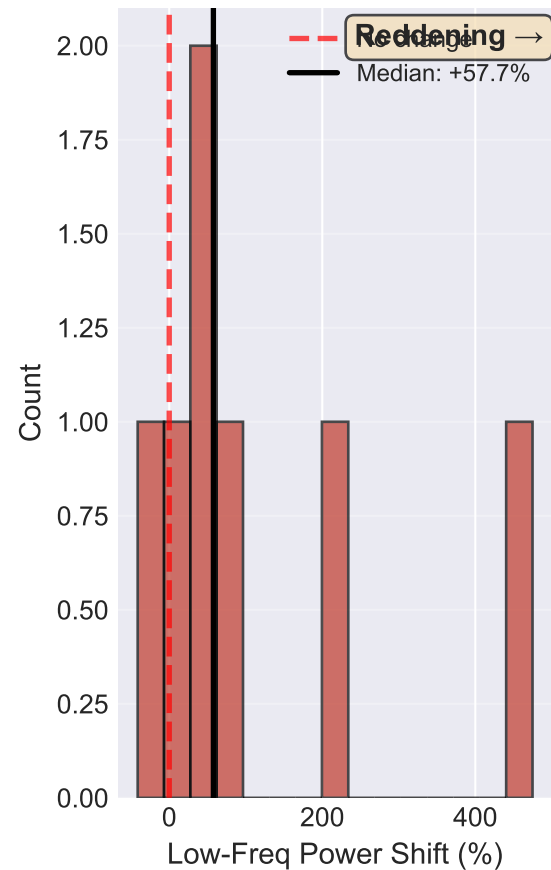


Low freq shift: +0.008(+21.5%)
% outbreaks slowing: 23.4%

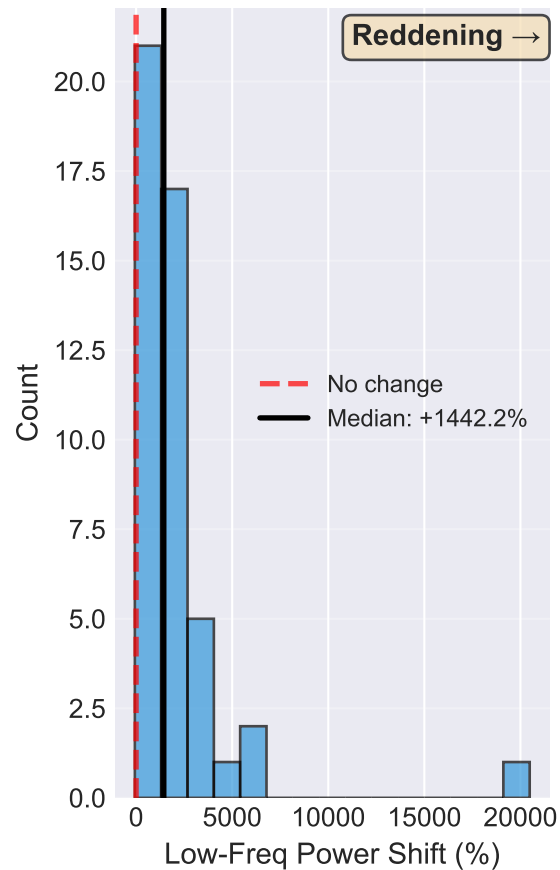
COVID U.S. (N=2806)



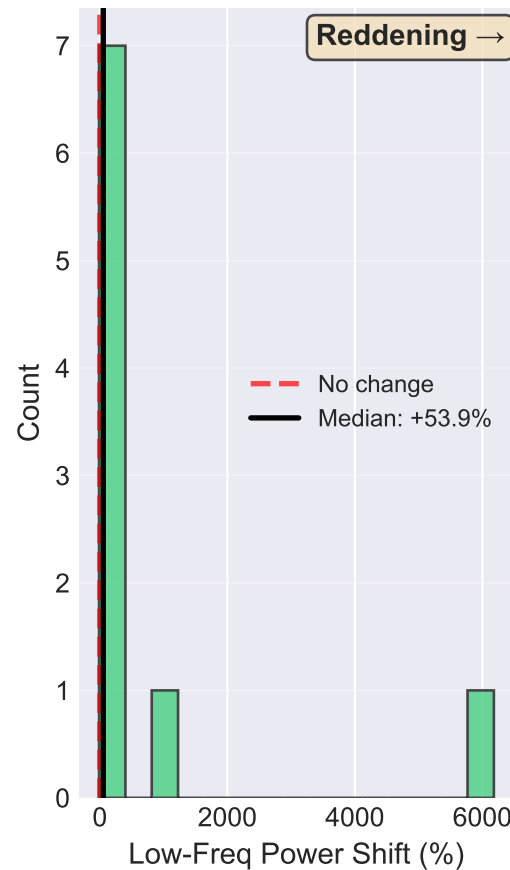
COVID



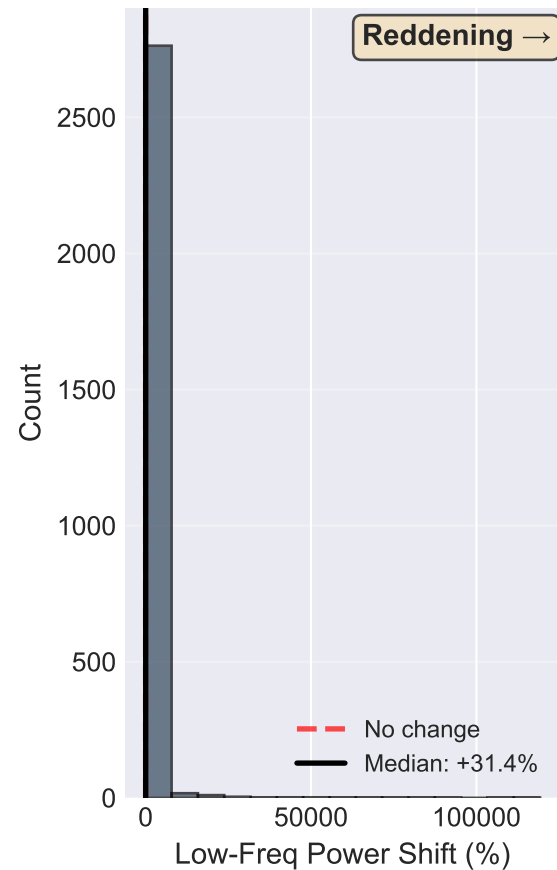
Mpox



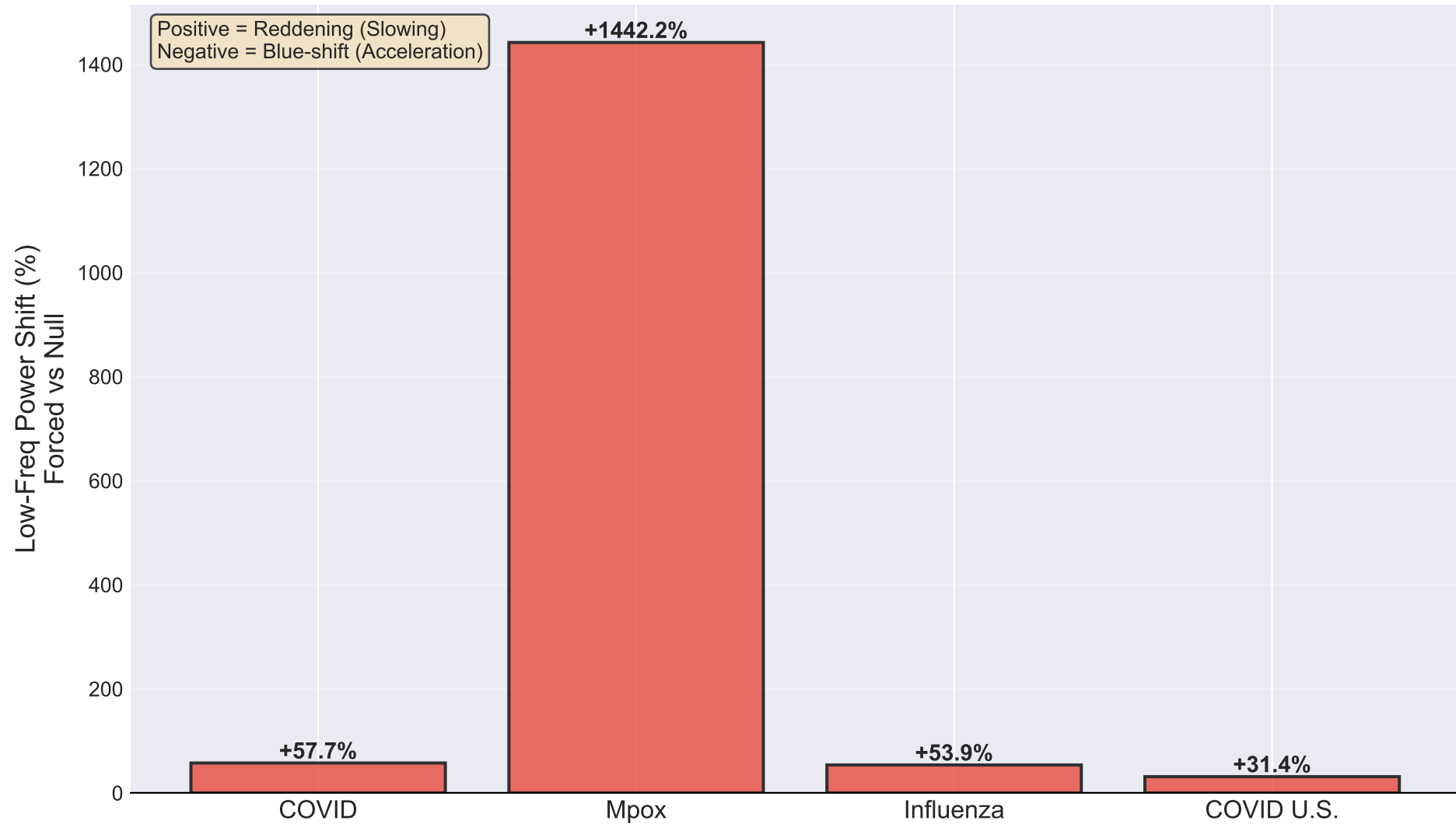
Influenza



COVID U.S.



Spectral Shift: Last 4 Weeks Approaching Bifurcation



Metric	COVID	Mpox	Influenza	COVID U.S.
Dominant Period (Null)	6.333 ± 1.795	2.438 ± 5.468	20.000 ± 7.892	3.462 ± 2.178
Dominant Period (Forced)	7.000 ± 0.817	7.111 ± 21.098	15.800 ± 5.267	6.374 ± 4.076
Dominant Period Shift	0.103 ± 1.832	4.622 ± 22.583	-0.083 ± 6.561	0.067 ± 4.344
Mean Frequency (Null)	0.252 ± 0.044	0.333 ± 0.061	0.126 ± 0.048	0.282 ± 0.035
Mean Frequency (Forced)	0.218 ± 0.034	0.202 ± 0.030	0.105 ± 0.018	0.262 ± 0.035
Mean Frequency Shift	-0.026 ± 0.025	-0.129 ± 0.068	-0.022 ± 0.038	-0.020 ± 0.029
Low Freq Power (Null)	0.018 ± 0.030	0.006 ± 0.033	0.152 ± 0.150	0.029 ± 0.032
Low Freq Power (Forced)	0.028 ± 0.035	0.056 ± 0.099	0.259 ± 0.115	0.034 ± 0.037
Low Freq Shift	0.007 ± 0.029	0.050 ± 0.101	0.091 ± 0.076	0.007 ± 0.042
Mid Freq Power (Null)	N/A	N/A	N/A	N/A
Mid Freq Power (Forced)	N/A	N/A	N/A	N/A
Mid Freq Shift	0.033 ± 0.065	0.324 ± 0.186	-0.040 ± 0.063	0.055 ± 0.094
High Freq Power (Null)	0.341 ± 0.153	0.592 ± 0.199	0.082 ± 0.103	0.424 ± 0.144
High Freq Power (Forced)	0.270 ± 0.136	0.168 ± 0.058	0.051 ± 0.015	0.355 ± 0.124
High Freq Shift	-0.039 ± 0.069	-0.407 ± 0.205	-0.031 ± 0.095	-0.063 ± 0.102
Spectral Entropy (Null)	3.626 ± 0.295	3.489 ± 0.504	3.631 ± 0.350	3.496 ± 0.428
Spectral Entropy (Forced)	4.276 ± 0.275	4.421 ± 0.292	3.769 ± 0.294	4.236 ± 0.359
Entropy Shift	0.644 ± 0.187	0.920 ± 0.401	0.221 ± 0.218	0.705 ± 0.264
Peak Sharpness (Null)	3.634 ± 1.238	3.225 ± 1.637	4.809 ± 3.618	3.845 ± 1.940
Peak Sharpness (Forced)	5.627 ± 2.321	4.555 ± 1.946	10.848 ± 2.661	5.372 ± 2.641
Peak Sharpness Shift	1.060 ± 1.353	1.209 ± 1.652	2.983 ± 2.434	1.242 ± 1.761
Reddening Index	0.111 ± 0.106	0.633 ± 0.331	0.144 ± 0.197	0.093 ± 0.150
Multiscale Index	-0.325 ± 1.519	-0.356 ± 1.974	-2.685 ± 2.586	-0.526 ± 1.924

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